Smart Gate v3: Change Register

1. Purpose

This register is the chain of custody for Smart Gate v3 (Objective O6). It records the nature, author and date of every significant change to the inherited system, why it was made, what it affected, where the original is kept, and who agreed it. Commit history in the repository holds line-level detail; this register is the readable summary.

- **Team repository:** https://github.com/martin213214/SmartGate-2026 (branch `main`)
- **Inherited baseline:** https://github.com/Latzerni/SmartGate.git, branch `main`, commit `2ea2c13385a71dbccf2d8aa182a95a092b86fd82` (as recorded in Appendix D). Original upstream: https://github.com/TheOpenSI/SmartGate

2. How to add an entry

1. Give the change the next ID (CR-nn).
2. Record the date, what changed, who made it and why.
3. List every file, part or setting affected and where the original is kept (for code, a backup copy or the commit hash).
4. Record who agreed it: OpenSI for scope or behaviour changes, the team for build-level choices within an agreed design.
5. Link the commit or document revision.
6. Back up the original **before** making the change.

3. Register

ID	Date	Change	Author	Reason	Affected	Original kept	Agreed by
CR-01	10 Sep	First L298N driver replaced	Janvy	Inconsistent output on ENB in Stage 1 testing	Motor driver	Faulty board set aside	Team
CR-02	14 Oct	Docker environment built (CUDA 13.2, OpenCV with GStreamer)	Martin Benny	Run the inherited application on the Orin	`Dockerfile`, `Makefile`, `docker/apt/`; `smartgate` image; `sgate` alias in `~/.bashrc`	n/a (new)	Team
CR-03	16 Oct	TensorRT engine rebuilt (10.16.2) from inherited YOLOv5 weights	Martin Benny	Inherited engine only runs on the device and version it was built for	`models/marsupial_16s.engine`, `.onnx`, `.onnx.data` (committed)	Inherited Jetson Nano engine in the baseline commit	Team
CR-04	1 Oct	Camera overlay set for IMX219 (CAM1) through jetson-io	Antus Jojo	Camera not detected on the Orin	`/boot/extlinux/extlinux.conf` overlay entry (on the Nano, not in repo)	Original `primary` boot entry left in `extlinux.conf`	Team
CR-05	20 Oct	Motor-pin register fixes identified (pins 16, 18, 29, 31)	Theartha Hari Nair	Orin header pins boot as inputs	Pin multiplexer registers `0x243D020`, `0x243D010`, `0x2430068`, `0x2430070`	n/a (runtime setting)	Team

ID	Date	Change	Author	Reason	Affected	Original kept	Agreed by
CR-06	22 Oct	Boot script applies the motor-pin fixes at start-up	Antus Jojo	Pin settings reset on every restart	`setup/sgate_boot.sh`	n/a (new)	Team
CR-07	20 Oct	Hall sensor supply set to 5 V	Theartha Hari Nair	Sensors need about 4.5 V; 5 V keeps output at a safe 3.34 V	Hall wiring	REV 0.3 diagram	Team
CR-08	15 Oct	Gate demonstration changed from LEDs to stepper motor with Hall end stops	Team	More representative test of the actuation chain	Hardware design, Deliverable 2	LED design in the proposal	Suggested by Alex (OpenSI)
CR-09	22 Oct	Stepper control merged into the existing IO and door controllers	Martin Benny	Drive the stepper and stop at the Hall sensors	`src/main/io_control.py`, `src/main/door_control.py`, `src/main/stepper_test.py` (new)	`io_control_dc.py`, `door_control_dc.py` (original DC-motor code); `io_control_old.py`, `door_control_old.py` (earlier stepper version)	Team
CR-10	27 Oct	Detection threshold raised from 0.5 to 0.6	Team	A person was read as Fallow Deer and closed the gate	`config/config.json`	0.5 value recorded here	OpenSI
CR-11	17 Sep	Test bench changed from an enclosed case to an open two-deck design	Martin Benny, Antus Jojo	Parts and wiring must be visible and changeable	Bench CAD and print (`docs/cad/`)	Version 1 CAD files, held by Martin Benny	OpenSI
CR-12	20 Oct	Bench assembly: Velcro mounting and partial hot-gluing	Team	Hole grid too large; printed joints fragile	Bench assembly	Removable (Velcro, hot glue)	Team
CR-13	25 Oct	Web interface extended into a dashboard: Dashboard page (detection charts and log) and Live Stream page (live feed, Open/Close/Stop, Auto/Manual, sensor readings, analytics)	Martin Benny	Remote-monitoring stretch goal	`src/web-app/frontend/stats.html`, `gates.html`, `wwwroot/js/stats.js`, `live.js`, `controllers/jetson_live.js`	Original `src/main/http_server.py` and `src/web/` (in repo)	OpenSI
CR-14	28 Oct	Dashboard live status, address panel and smartgate.local name	Krishna Sharma	Easier access from other devices	`src/web-app/frontend/gates.html`, `wwwroot/js/live.js`; `smartgate.local` host name set on the Nano (not in repo)	n/a (new)	Team
CR-15	1 Nov	Wiring diagram revised to REV 0.4 (as built)	Theartha Hari Nair	Record the actual bench wiring	`docs/wiring/`	REV 0.1–0.3, held by Theartha Hari Nair	Team
CR-16	24 Oct	Boot service rebuilt and setup script updated to include the pinmux step and live-code mount	Antus Jojo	Re-running the inherited setup script overwrote the working service and the gate stopped moving	`setup/systemd_service_setup.sh`; `/etc/systemd/system/smartgate.service` (on the Nano)	Inherited script in baseline commit	Team
CR-17	24 Oct	Web app platform (`src/web-app`): Node.js and PostgreSQL in Docker, with dummy roles, Alerts pages	Martin Benny,	Remote monitoring with user accounts and stored records (stretch goal)	`src/web-app ` (folder name has a trailing space, see note 3)	Inherited web app in the baseline commit	OpenSI

ID	Date	Change	Author	Reason	Affected	Original kept	Agreed by
			Krishna Sharma				
CR-18	31 Oct – 1 Nov	Code pushed to the team repo as a single clean commit; `.gitignore` added for large files	Antus Jojo	Push failed: old history held `yolov5/`, a venv and a 308 MB torch wheel over GitHub's 100 MB limit	Whole repo history; `.gitignore`	Full working copy on the Nano at `/home/smartgate/SmartGate`; inherited history upstream (baseline above)	Team
CR-19	1 Nov	README rewritten; change register, wiring, testing, bench design, build log and operating guide added to the repo	Antus Jojo, Theartha Hari Nair	Handover documentation (Objective O6)	`README.md`, `CHANGE_REGISTER.docx`, `docs/`	Inherited README in baseline commit	Team

4. Notes

1. **Controller backups.** The working notes refer to `io_control.bak` and `door_control.bak`, but the repo holds them as `io_control_dc.py` and `door_control_dc.py` (the original DC-motor code) and `io_control_old.py` and `door_control_old.py` (an earlier stepper-only version). DC mode is also still built into the current `io_control.py` (`MOTOR_MODE = 'dc'`, ENB on pin 12).
2. **Dashboard location.** The dashboard (CR-13, CR-14, CR-17) is the web app in `src/web-app`, on port 8000; the Dashboard page is `stats.html` and the Live Stream page is `gates.html`. The `smartgate.local` name is set on the Nano itself, not in the repo.
3. **Web app folder name.** `src/web-app` is committed as `src/web-app ` (trailing space). Fix with `git mv "src/web-app " src/web-app` and record it here as a new entry.

5. Items deliberately left unchanged

Item	Reason
YOLOv5 weights and 16-species class list	Out of scope; changing them would break comparison with OpenSI's instance
OPEN and CLOSE species lists	Ecological decision for OpenSI
Original web server code	Kept alongside the dashboard (`src/main/http_server.py`, `src/web/`)